Applying effective environmental forecasts to Management Strategy Evaluation of yellowtail flounder

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1 ABSTRACT

2 INTRODUCTION

As the world's oceans are complex physical and chemical systems, climate change has impacted oceans in a variety of ways such increased ocean temperature and acidification, changes in ocean currents, (Fabry et al., 2008; Guinotte & Fabry, 2008; Abraham et al., 2013; Sydeman et al., 2015; Wilson et al., 2016; Poloczanska et al., 2016; García Molinos et al., 2017; Friedland et al., 2022; Cheng et al., 2024). As a result, different marine species are projected to have complex and different responses to these environmental shifts depending on species' life histories, thermal tolerances, mobility, and other aspects of the species' biology. (Guinotte & Fabry 2008; Chown et al 2010; Honisch et al 2012; Stewart et al 2014; Sydeman et al 2015; Polocanska et al 2016; Wilson et al 2016; Chaikin et al 2024). This creates compounding challenges when managing commercially-important fisheries. As some species will shift ranges northward or to deeper waters in search of colder temperatures, this will disrupt historical food chains as predators and prey may shift in different directions, or range shifts will introduce new interactions between predator and prey species that have historically not overlapped (Overholtz et al 2009; Albout et al 2014; Stewart et al 2014; Sydeman et al 2015; Wilson et al 2016; Garcia Molinos et al 2017; Chaikin et al 2024). Further, as fish populations may experience distribution shifts, annual growth patterns, or vary in abundance, this has potential to increase catch effort as historical fishing grounds and times may now not line up with fish stocks' new life history dynamics (Overholtz et al 2009; Poloczanska et al 2016; Tommasi 2017; Fulton et al 2021). New species may also shift their ranges into a commonly fished area, potentially opening up new commercial opportunities (Stewart et al., 2014).

- THIS IS WHAT HAPPENS IF THE KEY DOESN'T EXIST IN THE .BIB DOC: (**IfYouCiteAKeyThatDoesntExist?**)

- TO PUT A CITATION IN THE MIDDLE OF A SENTENCE, Sydeman et al. (2015) SHOWS YOU CAN JUST NOT INCLUDE THE BRACKETS
- NOTICE HOW FORGETTING ONE @ SYMBOL OR BRACKET OR SEMI-COLON MESSES UP THE CITATION: [Fabry et al. (2008); Guinotte & Fabry (2008); abrahamReviewGlobalOcean2013; Sydeman et al. (2015); Wilson et al. (2016); Poloczanska et al. (2016); García Molinos et al. (2017); Friedland et al. (2022); Cheng et al. (2024)]

The US Northeast continental shelf has warmed faster than any other marine ecosystem in the United States (Saba et al 2016; Kleisner et al 2017; Saba et al 2023). The varying responses to climate change across fished species has presented challenges in accurately assessing stock health of various species in the region (Hare et al 2016; Kleisner et al 2017). Environmental covariates have rarely been incorporated into stock assessments in the Northeast (Mazure et al 2023). This has shown to reduce retrospective patterns and increase projection accuracy in New England populations such as Atlantic Cod, haddock, yellowtail flounder, and winter flounder (Buckely et al 2004; Klein et al 2017).

Management Strategy Evaluations (MSEs) are decision making frameworks that allow assessors to simulate and compare the sustainability and tradeoffs associated with different management decisions (Tommasi 2017; Mazure et al 2023; Saba et al 2023). Applying climate change covariates and MSE to new England groundfish has shown to improve the robustness of stock assessments (Mazur 2023). However, applications of MSEs can still be challenging, as climate change shifts population baselines and creates dynamics that are harder to predict in biological systems. This demonstrates a need for further research on how to incorporate these environmental conditions into the MSE framework (Hollowed et al 2011; Punt 2016; Holsman et al 2022; Saba et al 2023). Populations' responses to climate change can be complex, as fish exhibit different sensitivities to environmental conditions depending on their lifestage, and different environmental changes (i.e. increased sea surface or bottom temperature, changes in salinity, and altered currents) will have different effects on populations (Hollowed et al 2011).

Further, decisions on how to project environmental conditions are important to consider when predicting stock dynamics. As assessments attempt to capture future population trends, forecasting environmental trends is also imperative, however this can be challenging as optimal decisions can change depending on scale, species, and which environmental covariates are used (Hollowed et al 2011; Tommasi 2017; Holsman et al 202).

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